MIL-B-52301C
23 January 1984
SUPERSEDING
MIL-B-52301B
2 March 1970

MILITARY SPECIFICATION

BRACKET ASSEMBLY, CHAIN HOIST SUPPORT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a bracket assembly and its components to be used with a trestle assembly (MIL-T-52150).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents. The following other Government documents, form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13207E9734	- Bracket Assembly, Chain Hoist Support.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS A5.1	- Covered Carbon Steel Arc Welding Electrodes.
ANSI/AWS A5.18	- Carbon Steel Filler Metals for Gas Shielded Arc Welding.
ANSI/AWS D1.1	- Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The bracket assembly or components shall be as shown on Top Assembly TA13207E9734 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the bracket assembly. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

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3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Material deterioration and control. The bracket assembly shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the bracket assembly may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Welding electrodes. Welding electrodes shall conform to ANSI/AWS A5.1 or ANSI/AWS A5.18, classification and size as required.

3.5 Identification marking. The bracket assembly or components, as applicable, shall be metal die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Treatment and painting. The bracket assembly or components, as applicable, with the exception of the pin assembly and the bridge pin retainer assembly, shall be cleaned, treated,and painted in accordance with MIL-T-704, type B.

3.7 Stenciling. The nomenclature, National stock number, weight, and cube shall be stenciled on each bracket assembly. The letters "NET WT" shall precede the numerals in the weight and letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 1/2 inch high. Stenciling shall conform to MIL-T-704.

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3.8 Workmanship.

3.8.1 Steel fabrication. Steel may be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove cooling checks. Holes shall be drilled to full size or drilled undersize and reamed to full size through steel templates. All burrs shall be removed.

3.8.2 Welders and welding.

3.8.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determing that automatic welding equipment operators are capable of producing quality welds in accordance with ASME or AWS codes.

3.8.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the ANSI/AWS D1.1 - Structural Welding Code. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wirebrushed clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the

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right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:.

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article bracket assembly or components, as applicable, shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The bracket assembly or components, as applicable, shall be examined as specified herein for the following defects:

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Major

101. Material not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Components missing or not as specified.
106. Workmanship not as specified.
107. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.
108. Any part or component not in accordance with the QAP requirements as shown on the drawings.

Minor

201. Identification marking incorrect, missing, or illegible.
202. Treatment and painting not as specified.
203. Stenciling not as specified.
204. Dimensions other than those identified by QAP data not as specified.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

109. Containers not as specified for level A or B. Each incorrect container shall constitute one defect.
110. Components not secured within the boxes to prevent free movement as specified for level A or B.
111. Strapping not as specified for level A.
112. Gross weight of wood boxes exceeds 250 pounds for level A or B.

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113. Gross weight or size of fiberboard boxes exceeds the box limitation for level B.
114. Fiberboard box not sealed and reinforced as specified for level B.
115. Packing not in accordance with the referenced document as specified for commercial.
116. Marking missing, illegible, incorrect or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. The complete bracket assemblies shall be packed in close-fitting boxes conforming to PPP-B-621, class 2, style optional; or boxes conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed 250 pounds. Blocking and bracing to prevent movement and damage shall be provided in accordance with MIL-STD-1186. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.1.1.1 Pin and bridge pin retainer assemblies. When procured separately from the bracket assembly, the pin assemblies or bridge pin retainer assemblies shall be packed as specified in 5.1.1 for complete bracket assemblies.

5.1.2 Level B. The complete bracket assemblies shall be packed as specified for level A, except the boxes shall be domestic type or class as applicable. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.1.2.1 Pin and bridge pin retainer assemblies. Assemblies of like description, procured separately from the bracket assembly, shall be packed together in close-fitting boxes conforming to PPP-B-621, class 1, style optional; PPP-B-601, domestic type, style optional; or PPP-B-636, V3c, style optional. The contents shall be secured within the box in a manner to prevent free movement in accordance with MIL-STD-1186. The gross weight or size of fiberboard boxes shall not exceed the limitations of the box specification, and the boxes shall be waterproof sealed and reinforced with bands of reinforced tape in accordance with the appendix to the box specification. The gross weight of wood boxes shall not exceed 250 pounds and the box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.1.3 Commercial. The components comprising a complete bracket assembly, and the pin assembly and bridge pin retainer assembly, procured separately from the bracket assembly, shall be packed in accordance with MIL-STD-1188.

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5.2 Marking.

5.2.1 Military. Shipping containers for level A or B shall be marked in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The bracket assembly and its components are intended for use in floating and fixed bridges to support the chain hoist when the chain hoist is used to adjust the height of the transom in the trestle assembly.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2).

c. Level of packing required (see 5.1).

d. When other than finish B strapping is required (see 5.1.1).

6.3 First article. When a first article inspection is required the item shall be examined and should be a first produced bracket assembly. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and approval of the documents first article.

6.4 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

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6.5 Definition.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements Section 4 of the specification. QAP's indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Recycled materiel. It is encouraged that recycled materiel be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
Army - ME

Review activity:
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Preparing activity:
Army - ME

Project 5420-0148

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-52301C	Bracket Assembly, Chain Hoist Support

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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